Predicting Earned Run Average

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How Predictable Are Pitching Metrics?

- Many arguments state that a pitcher is only responsible for the “three true outcomes” (home runs, strikeouts, and walks)
  - BABIP (Batting Average for Balls in Play) is highly variable from year to year, and dependent on the fielders
  - But can the pitcher be graded on quality of contact (e.g., barrels, launch angle, launch velocity)?
- How much of a pitcher’s results are statistical noise (e.g. sampling effects) as opposed to fluctuations in true “quality”, age, injury, team defense, etc?

K%-BB% Is Strongest Predictor of Next Year's ERA

- Compared one year's metric to the next year's ERA
  - Used pitchers with more than 140 IP in each of successive year pairings anytime during 2021-2025 (380 year pairings)
- ERA is not a good predictor of next year's ERA
- K%-BB% is the best predictor

Correlation Coef. from Year to Next Year

	Correlation Coefficient
ERA	0.28
K%-BB%	0.40*
FIP	0.34
xFIP	0.37
SIERA	0.38
wOBA	0.32
xwOBA	0.35
xwOBAcon	0.11

* Only the absolute value of the correlation is relevant

Shouldn't a Metric Be Also Be Good for the Current Year?

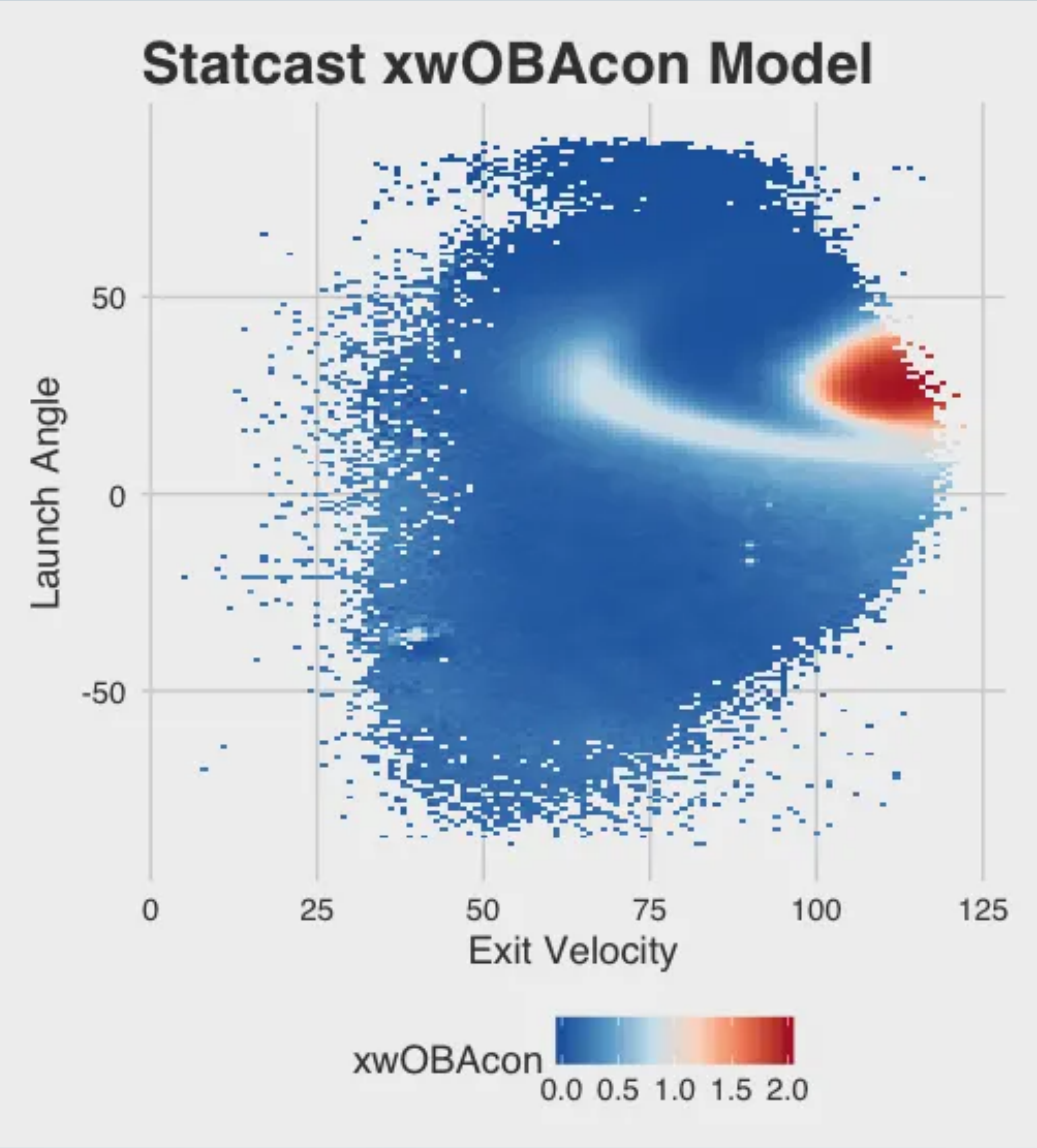
Metric Comparison from Year to Year

	Correlation to Next Year ERA	Correlation to Current Year ERA
ERA	0.28	
K%-BB%	0.40*	0.54*
FIP	0.34	0.79
xFIP	0.37	0.60
SIERA	0.38	0.61
wOBA	0.32	0.90
xwOBA	0.35	0.77
xwOBAcon	0.11	0.63

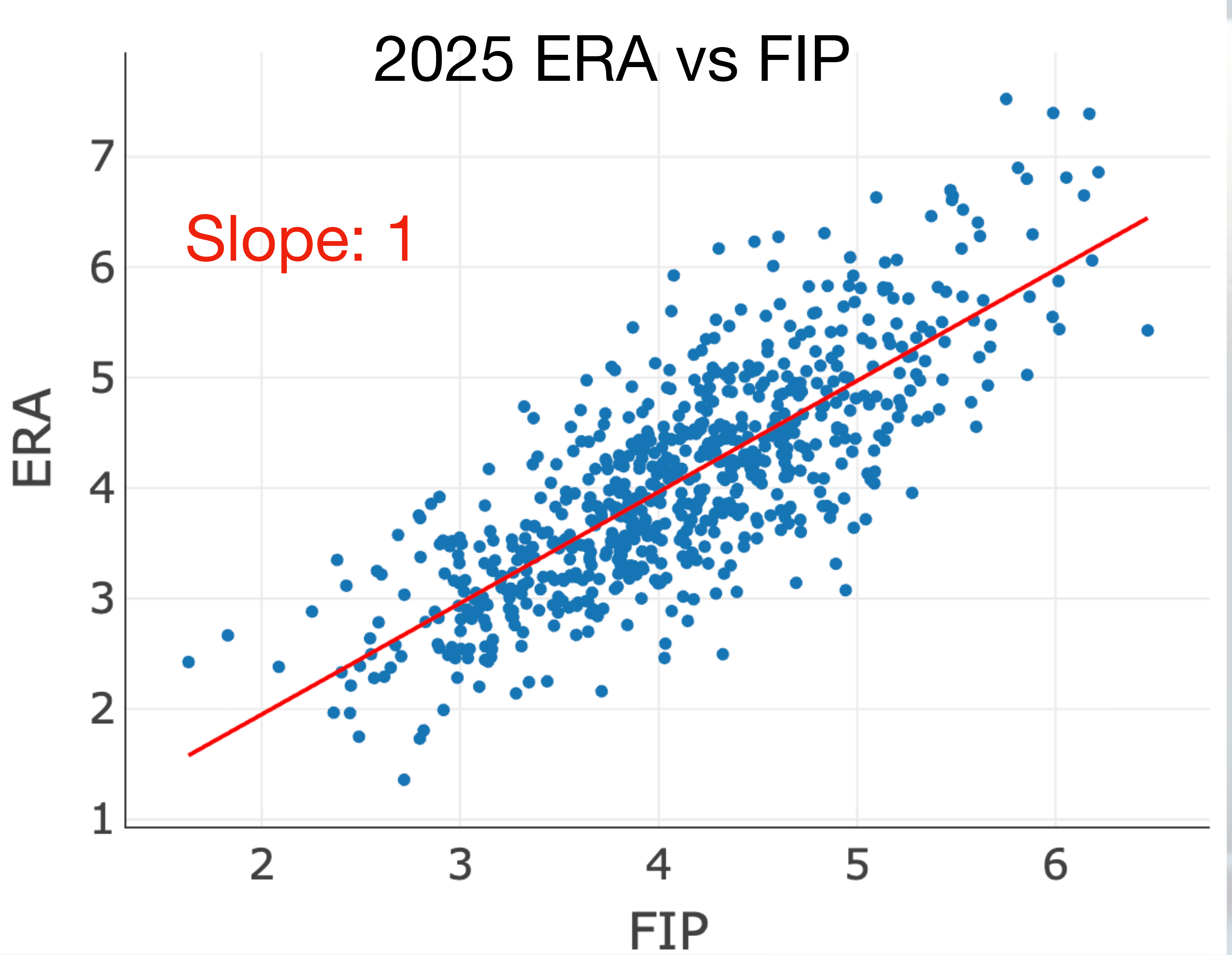
* Only the absolute value of the correlation is relevant

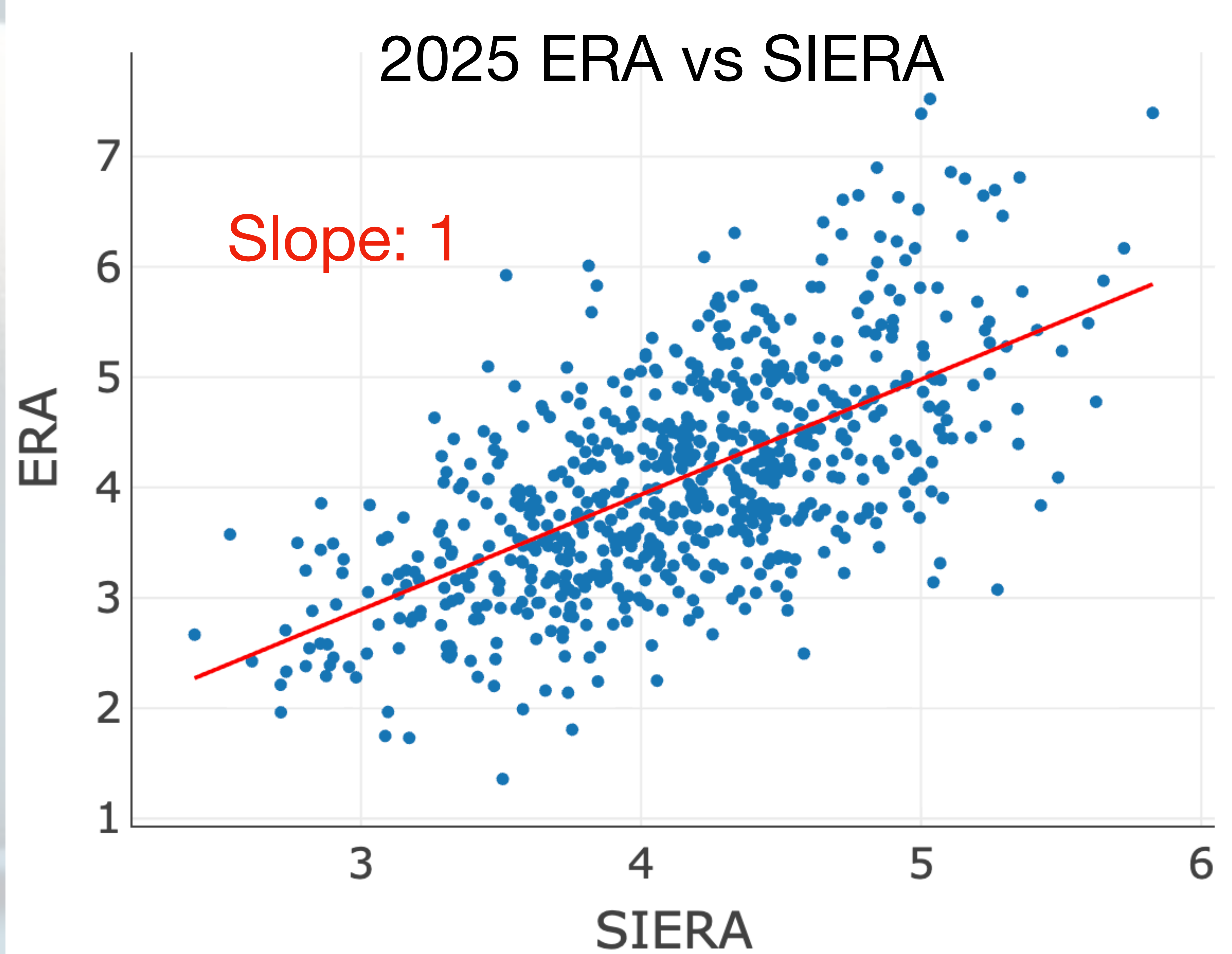
wOBA vs xwOBA

- xwOBA is the expected outcome for each plate appearance
  - Strikeouts are like any other out
- For batted balls, xwOBAcon (xwOBA contact) is the expected wOBA outcome of each launch angle/velocity pair
  - Likelihoods for 1B/2B/3B/HR/out times the coefficient for each
- xwOBA is the overall estimate for wOBA to include xwOBAcon, strikeouts, and walks



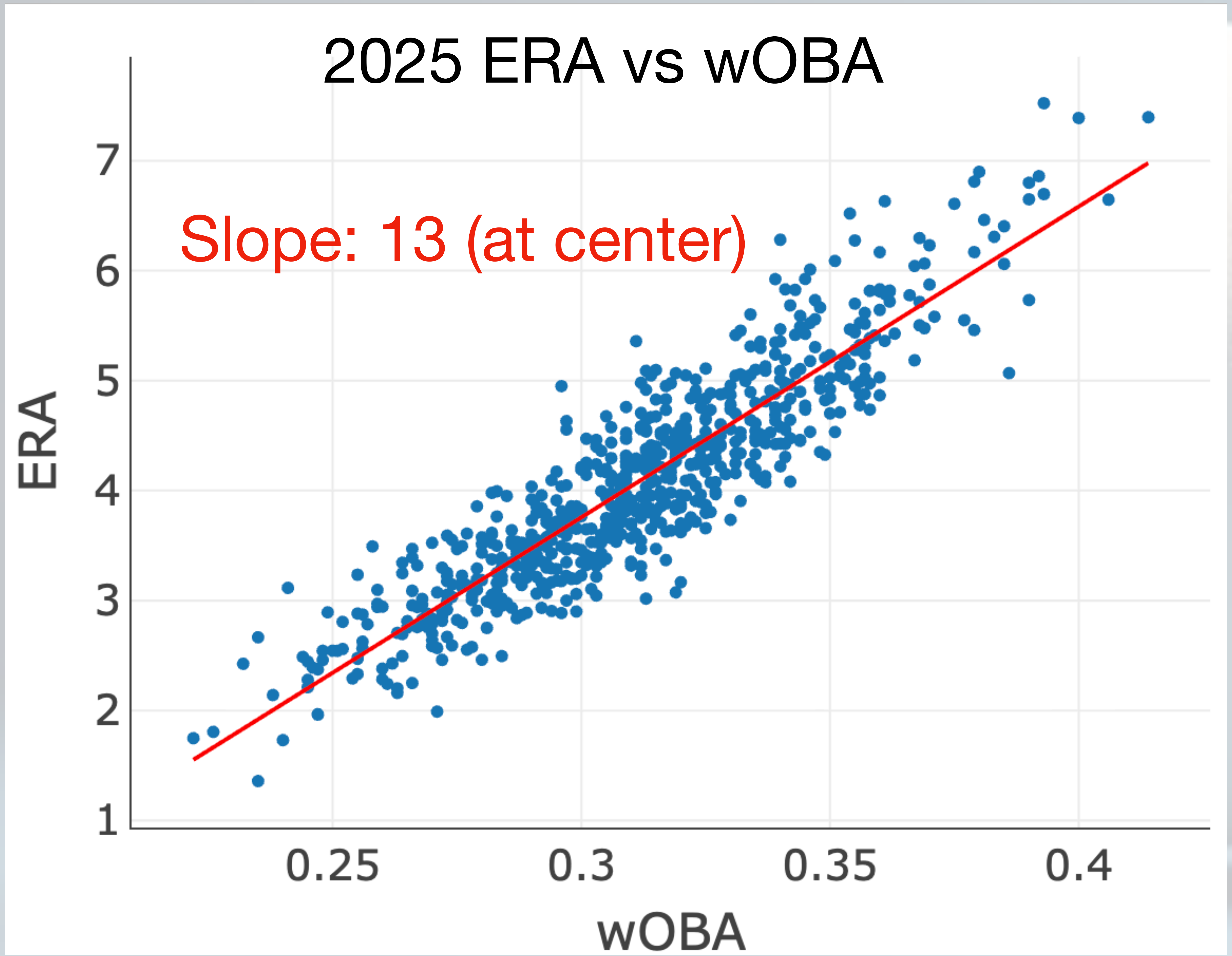
2025 Correlations of ERA vs FIP & SIERA

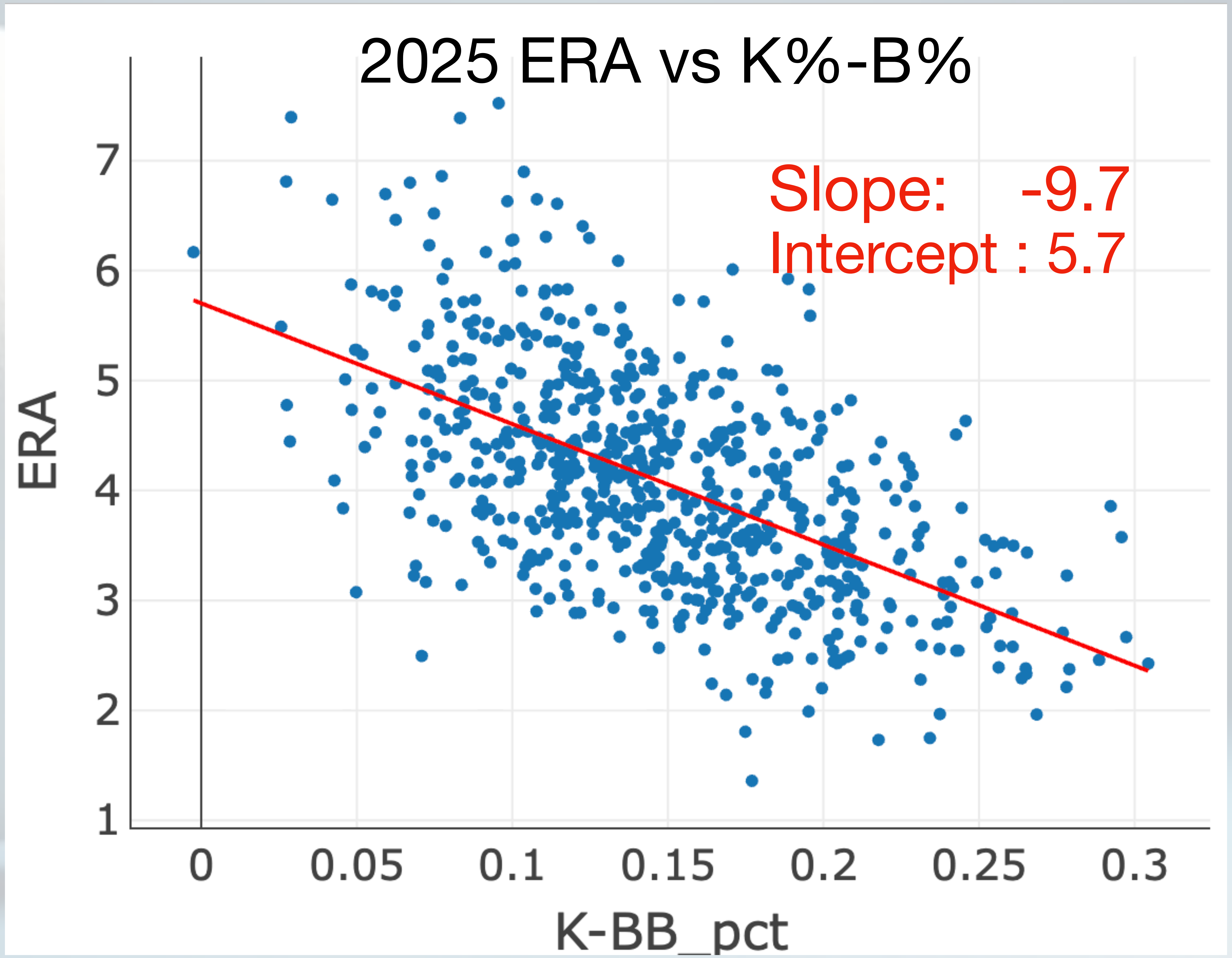




Both scaled the same as ERA by design

2025 Correlations of ERA vs wOBA & (K%-BB%)





Teaser: Slope of the line will impact the statistical variability

Variability Factors in Pitcher Outcomes

- “True” pitching merit
- Three true outcomes (K, BB, HR)
- Batted ball outcome (hit vs. out)
- Quality of contact (barrels, launch angle or velocity)
- Ordering of hits (variability in inning outcomes)
- Earned vs unearned runs (about 10% of total runs are “unearned”)
- Fielding (range, etc.)
- Park effects
- Statistical variance

Which Effects Are Included in Each Metric?

	ERA	K%-BB%	FIP	wOBA	xwOBA	SIERA
“True” pitching merit						
Three true outcomes		K, BB		BB, HR	BB,HR	
Batted ball outcome						
Quality of contact						GB, FB
Ordering of events						
Earned & unearned						
Fielding						
Park effects						
Statistical variance						

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Park effects						
Statistical variance						

The variance will be the sum of the variances for each of these terms

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Ordering of events						
Earned & unearned						
Fielding						
Park effects						
Statistical variance						

Can we freeze these and look at just the statistical variance of each estimator?

Hypothesize an “Ideal” League Average Pitcher

- Freeze all underlying likelihoods
  - I used league average values
- Random draws with same expected stats done over 100,000 ‘seasons’
- What does sampling do the resultant metrics?

Runs/Inning (2000-2025)

Runs	Innings	% of IP
0	157,231	74.69%
1	31,453	13.63%
2	14,894	6.45%
3	6,864	2.97%
4	3,000	1.30%
5	1,269	0.55%
6	559	0.24%
7	218	0.09%
8	86	0.04%
9	39	0.02%
10	18	0.01%
11	6	0.00%
12	1	0.00%
13	2	0.00%
14	1	0.00%

Outcome Frequency (2025)

	2025 Amount	2025 Pct/ PA
BB	15,379	8.4%
IBB	556	0.3%
HBP	1,928	1.1%
1B	26,115	14.3%
2B	7,745	4.2%
3B	628	0.3%
HR	5,650	3.1%
K	40,645	22.3%
PA	182,926	
IP	43,075	
lgERA	4.15	

Seasonal Results Change Based on Sampling

- If you toss a fair coin 100 times, you will only get exactly 50 heads about 8% of the time
- Each of the underlying stats (K%, BB%, HR%, xwOBA, etc.) have a sampling effect with a variance
- To baseline each, let's match them to their impact on an ERA estimate
- Parameter variance will scale by the square of the slope and the parameter weighting
- The overall variance of the sum of the component variances*

* (minus any cross-correlation effects)

ERA Distribution for a League-Average Pitcher

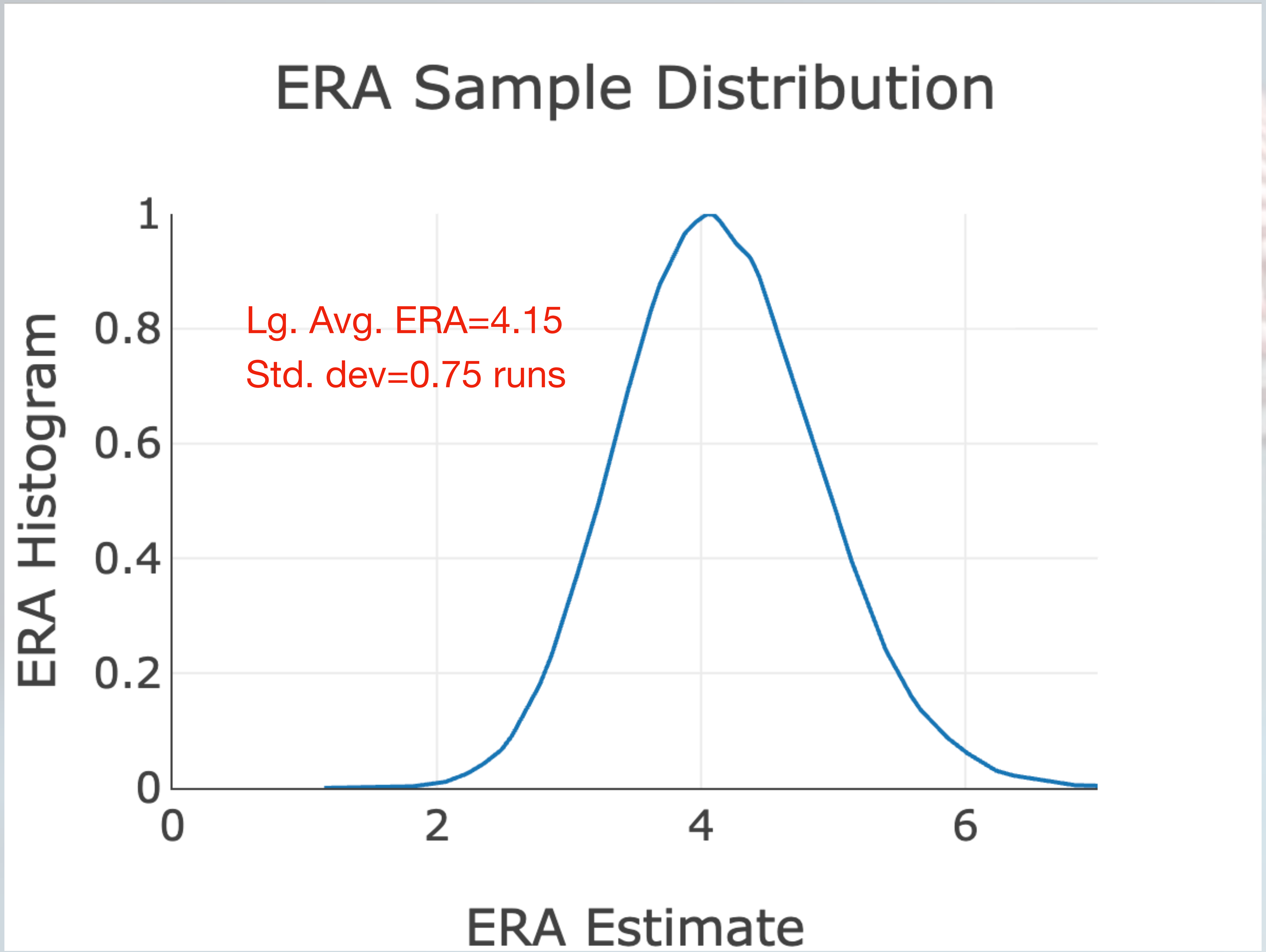
- 2025 league average ERA is 4.15
- Average earned runs per inning is
  $1p(1\ \ run) + 2p(2\ \ runs) + 3p(3\ \ runs) + \cdots$
- Average earned runs per game is nine times that or
  $9p(1\ \ run) + 18p(2\ \ runs) + 27p(3\ \ runs) + \cdots$
- Average over many 140 inning "seasons"
- Variance is the sum of all variances*
  - Standard deviation is square root of variance
- Standard deviation is 0.75 runs!

Runs/Inning (2000-2025)

Runs	% of IP	Var (140 IP)	Eff. Var
0	74.69%	0.0000	0.0000
1	13.63%	0.0008	0.0681
2	6.45%	0.0004	0.1397
3	2.97%	0.0002	0.1503
4	1.30%	0.0001	0.1188
5	0.55%	0.0000	0.0791
6	0.24%	0.0000	0.0503
7	0.09%	0.0000	0.0268
8	0.04%	0.0000	0.0138
9	0.02%	0.0000	0.0079
10	0.01%	0.0000	0.0045
11	0.00%	0.0000	0.0018
12	0.00%	0.0000	0.0004
13	0.00%	0.0000	0.0008
14	0.00%	0.0000	0.0005
Sum(var)			0.6629
Cross corr.			-0.0959
Total Var			0.5671

* (minus any cross-correlation effects)

ERA Distribution for a League-Average Pitcher



140 IP season over 100,000 trials

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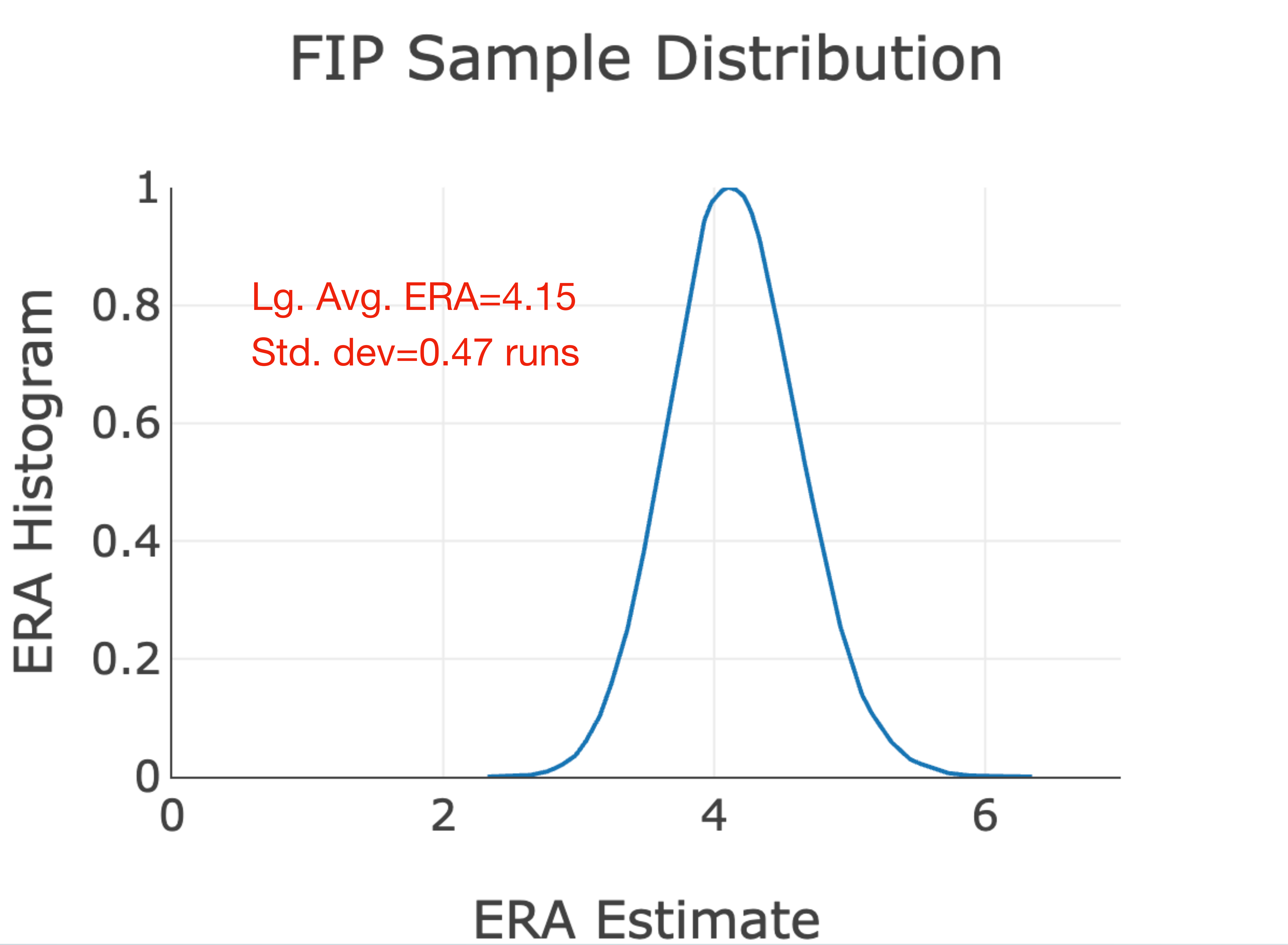
FIP (xFIP) ERA Estimate



630 TBF season (140 IP * 4.5 batters/IP) over 100,000 trials

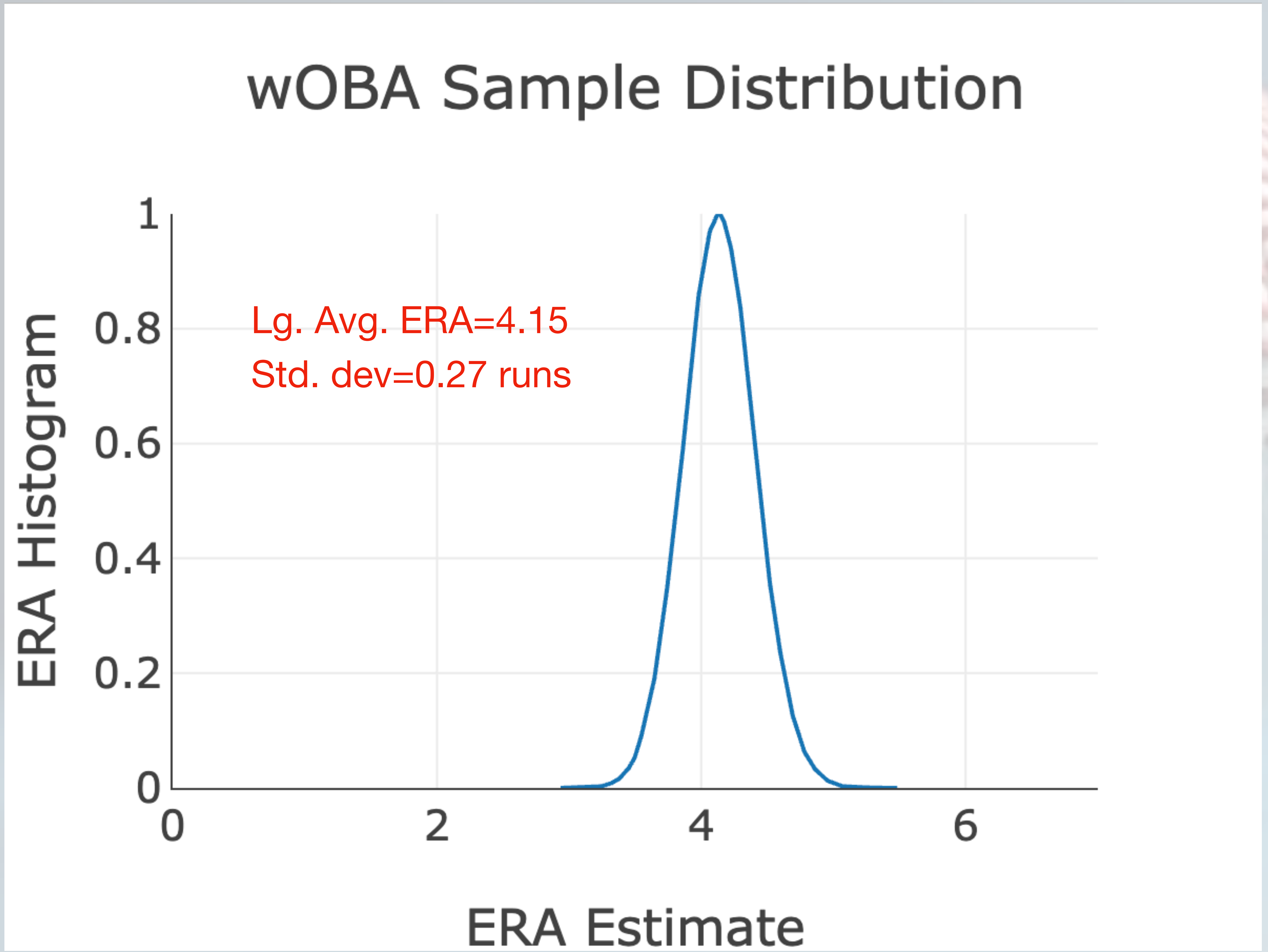
- 630 TBF season
  - 140 IP with 1.5 WHIP
- Slope multiplier is 4.5
  - Convert per AB stats to per inning

K%-BB%

Runs	Rate	Weight	Var (630 TBF)	Eff. Var
K%	22.30%	-2	0.0003	0.0223
BB%	8.40%	3	0.0001	0.0223
HBP%	1.06%	3	0.0000	0.0030
HR%	3.10%	13	0.0000	**0.1632**
Sum(var)				0.2107
Cross corr.				0.0119
Total Var				0.2226

HR% contribute to a high variance

wOBA (xwOBA) ERA Estimate



630 TBF season (140 IP * 4.5 batters/IP) over 100,000 trials

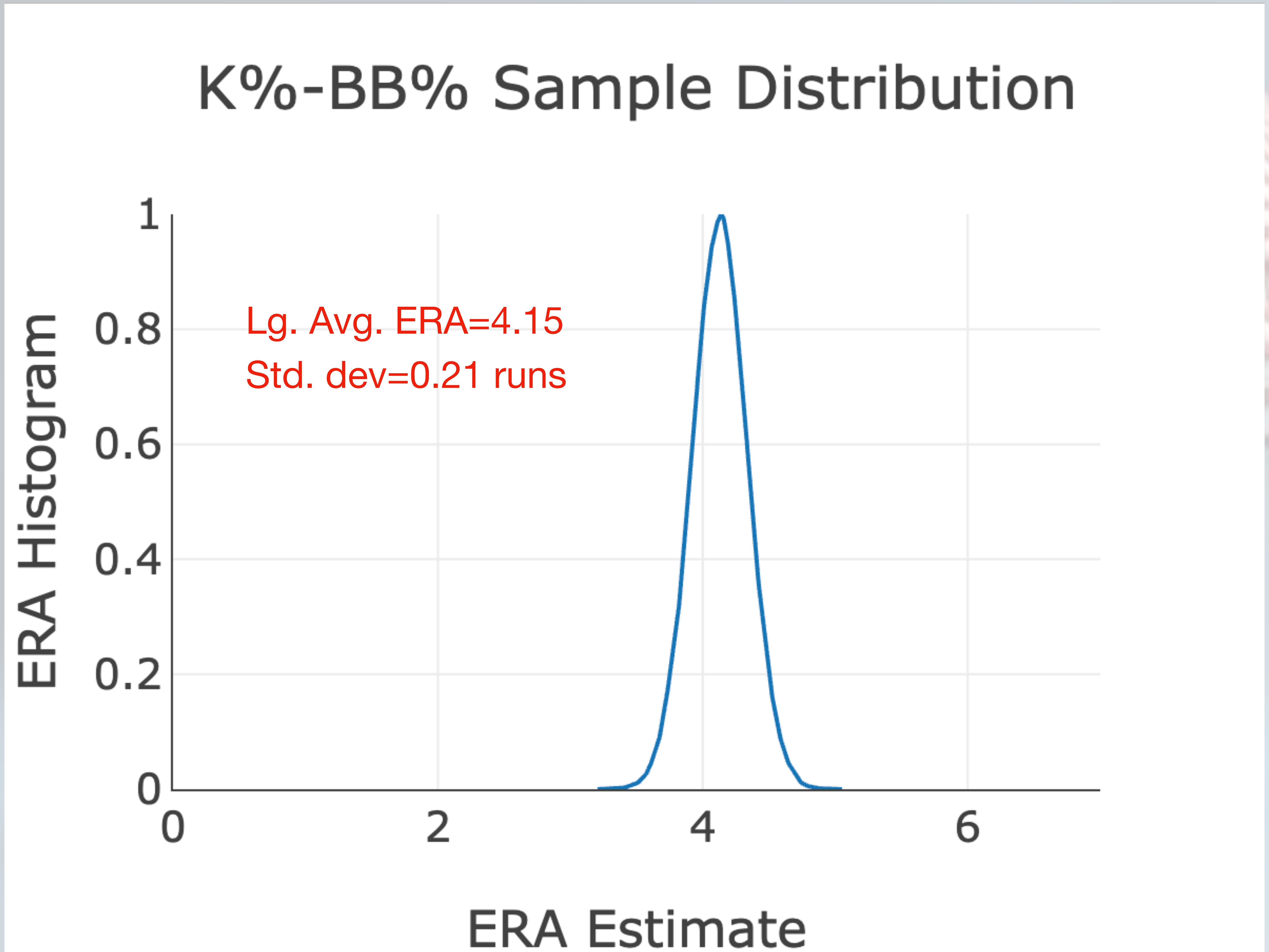
- 630 TBF season
  - 140 IP with 1.5 WHIP
- Slope multiplier is 13

wOBA (xwOBA)

Runs	Rate	Weight	Var (630 TBF)	Eff. Var
BB	8.40%	0.701	0.0001	0.0101
HBP	1.06%	0.733	0.0000	0.0015
1B	14.32%	0.895	0.0002	0.0264
2B	4.25%	1.270	0.0001	0.0176
3B	0.34%	1.607	0.0000	0.0023
HR	3.10%	2.066	0.0000	0.0344
Sum(var)				0.0924
Cross corr.				-0.0198
Total Var				0.0726

HR% scaled closer to other hits

K%-BB% ERA Estimate



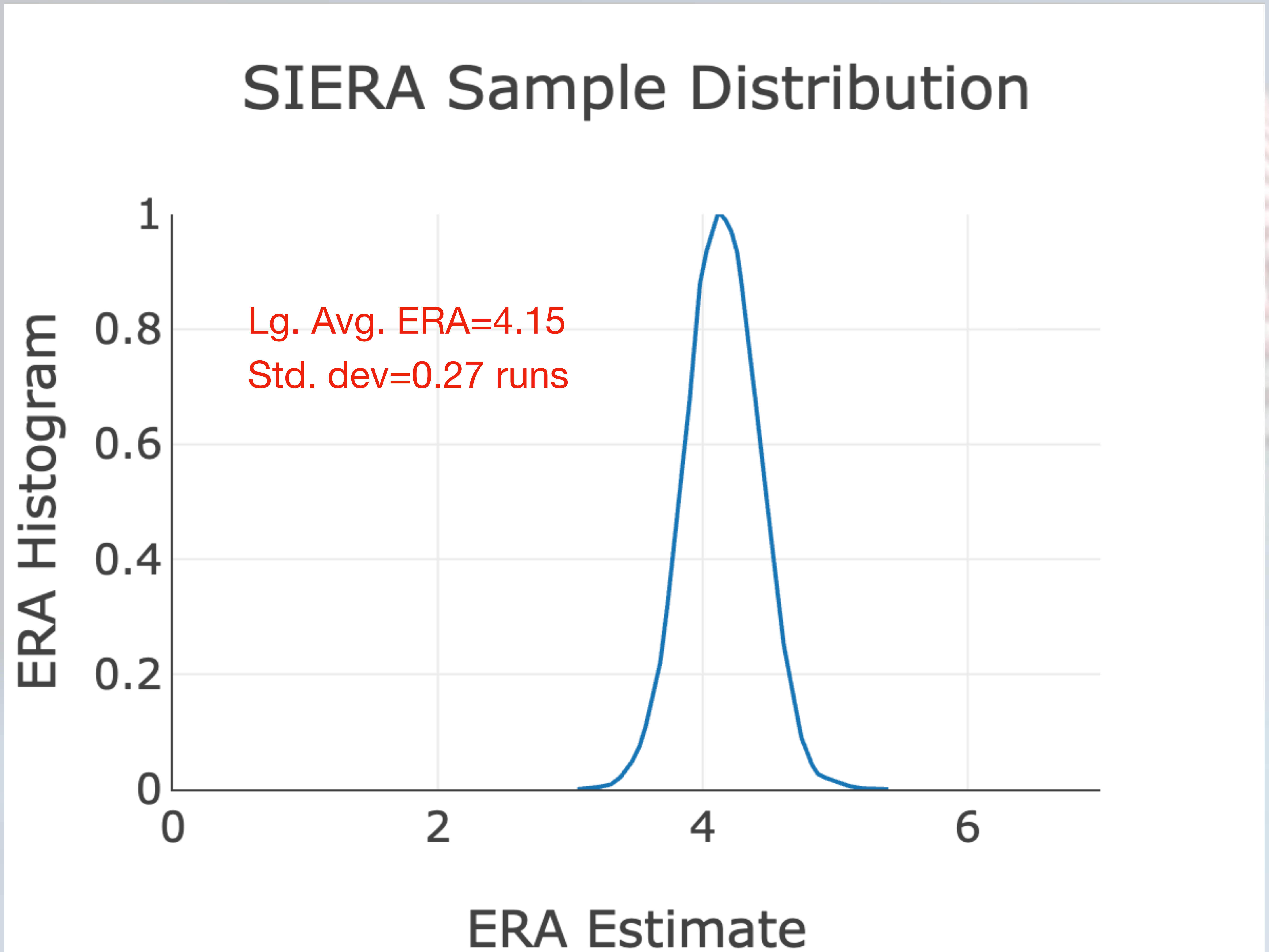
630 TBF season (140 IP * 4.5 batters/IP) over 100,000 trials

- 630 TBF season
  - 140 IP with 1.5 WHIP
- Slope multiplier is -9.7
  - slope of ERA/(K%-BB%)

K%-BB%

Runs	Rate	Var (630 TBF)	Eff. Var
K%	22.30%	0.0003	0.0259
BB%	8.40%	0.0001	0.0115
Sum(var)			0.0374
Cross corr.			0.0053
Total Var			0.0427

SIERA ERA Estimate

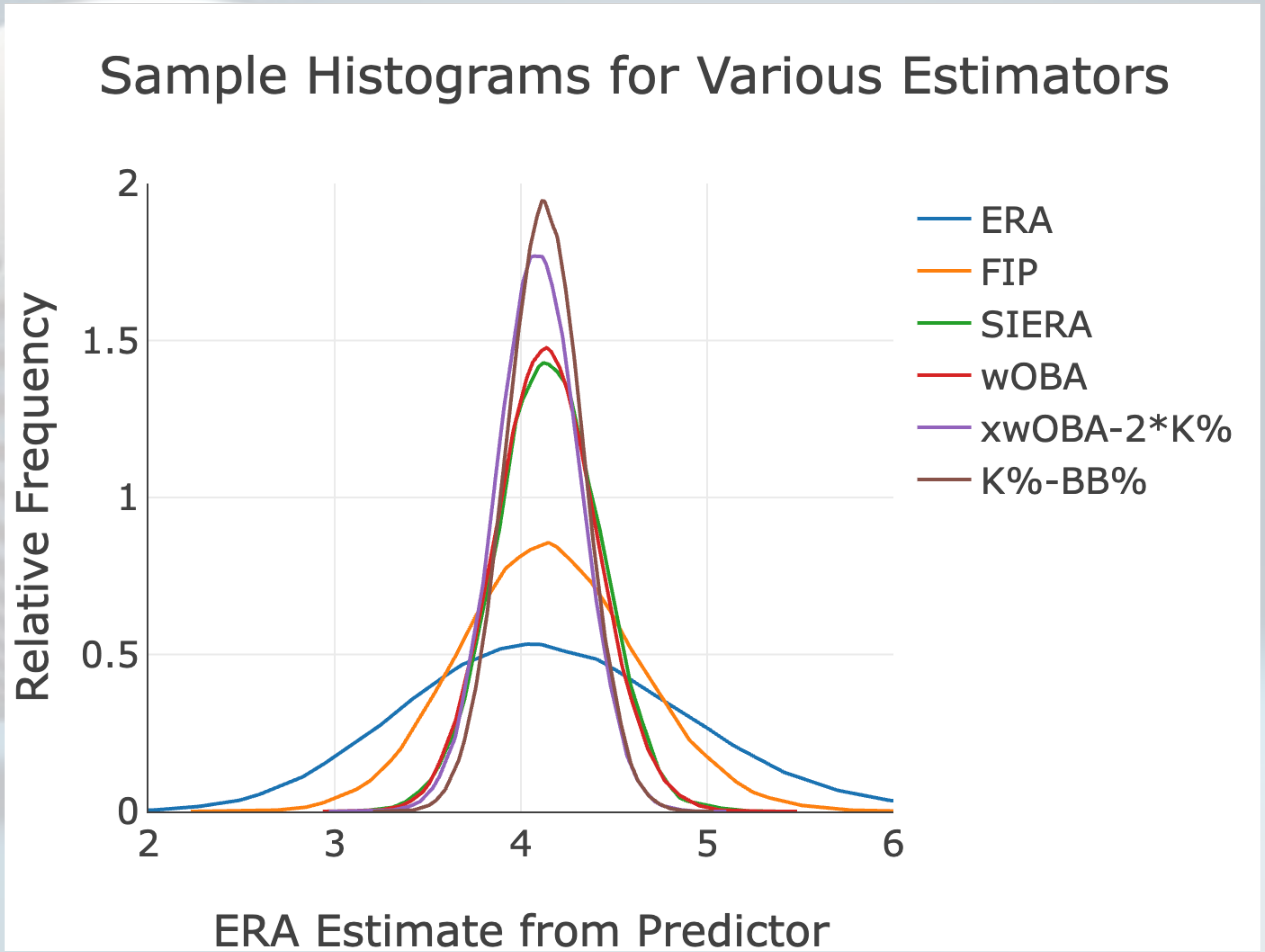


630 TBF season (140 IP * 4.5 batters/IP) over 100,000 trials

- 630 TBF season
  - 140 IP with 1.5 WHIP
- SIERA is based on SO, BB, ground ball rate, and fly ball rate
- SIERA includes non-linear effects, so the component variances are hard to isolate

Sampling Errors

- ERA has a large sample error (IP<TBF for starters)
- Correlating one noisy estimate (next year's ERA) with another noisy estimate will add the noise variances
- Goodness of fit should also account for sampling variances (e.g. that's why FIP does worse than K%-BB%)



Variance of the Sum of Random Discrete Variables

- A discrete random number with likelihood p has variance $p(1-p)$
- When taking the sum of n samples, the variance is then $p(1-p)/n$
- For a metric estimate of $\sum_i w_i p_i$, the variance is

$$\sigma^2 = \frac{1}{n} \sum_{i,j} C_{i,j}$$

where $C_{i,i} = w_i^2 p_i (1 - p_i)$ and $C_{i,j \neq i} = -\, w_i w_j p_i p_j$

Variance of Each Estimator: Summary

Variance and Future Correlation Coefficient of Estimators

Estimator	Sampling Std. Dev. (Runs)	Next-year Correlation Coefficient
ERA	0.75	0.28
K%-BB%	0.21	0.40
FIP (xFIP)	0.47	0.34
SIERA	0.27	0.37
wOBA (xwOBA)	0.27	0.38

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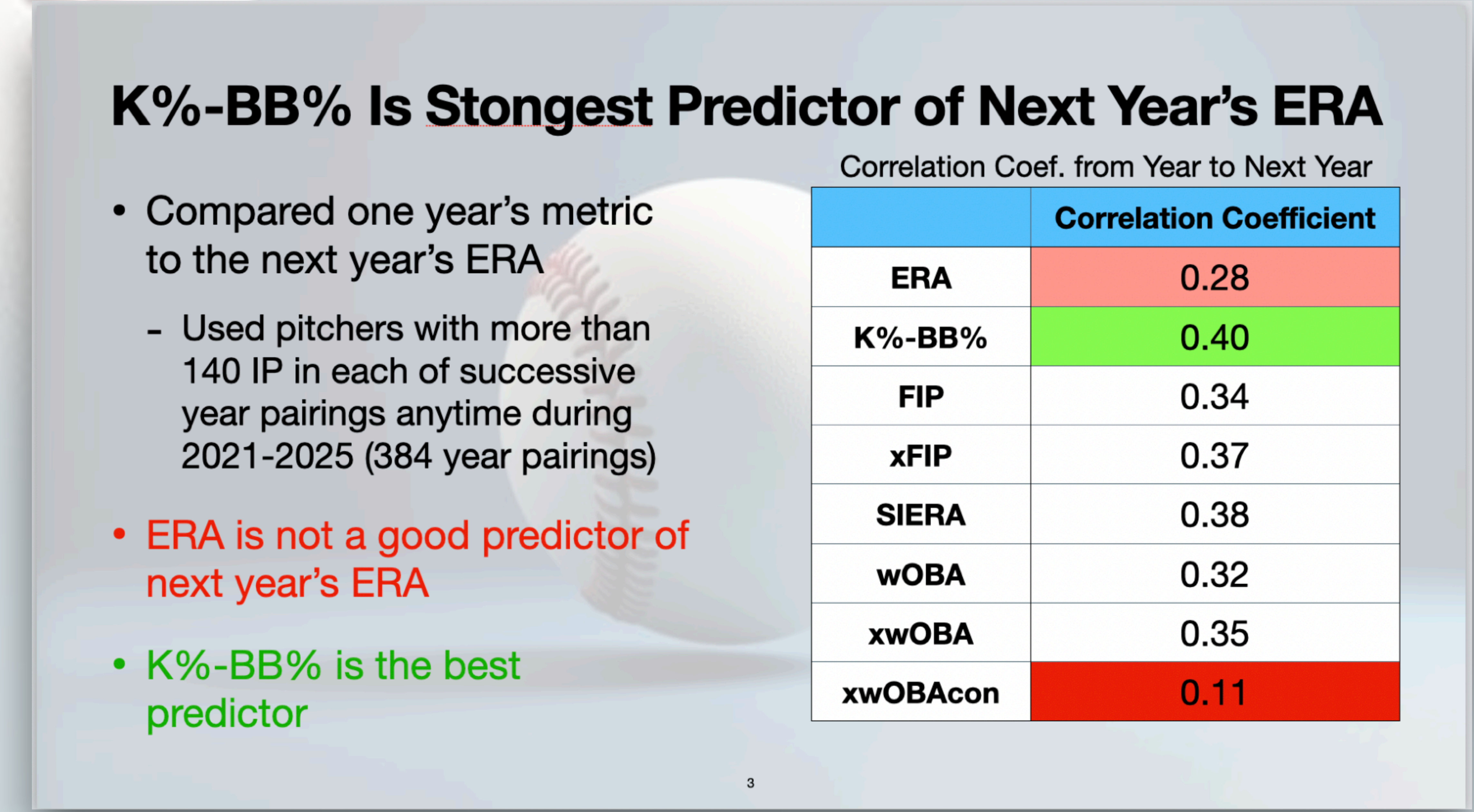

K%-BB% Is Stongest Predictor of Next Year's ERA

- Compared one year's metric to the next year's ERA
  - Used pitchers with more than 140 IP in each of successive year pairings anytime during 2021-2025 (384 year pairings)
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xwOBA-2*K%	0.24	0.42

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A better estimator?

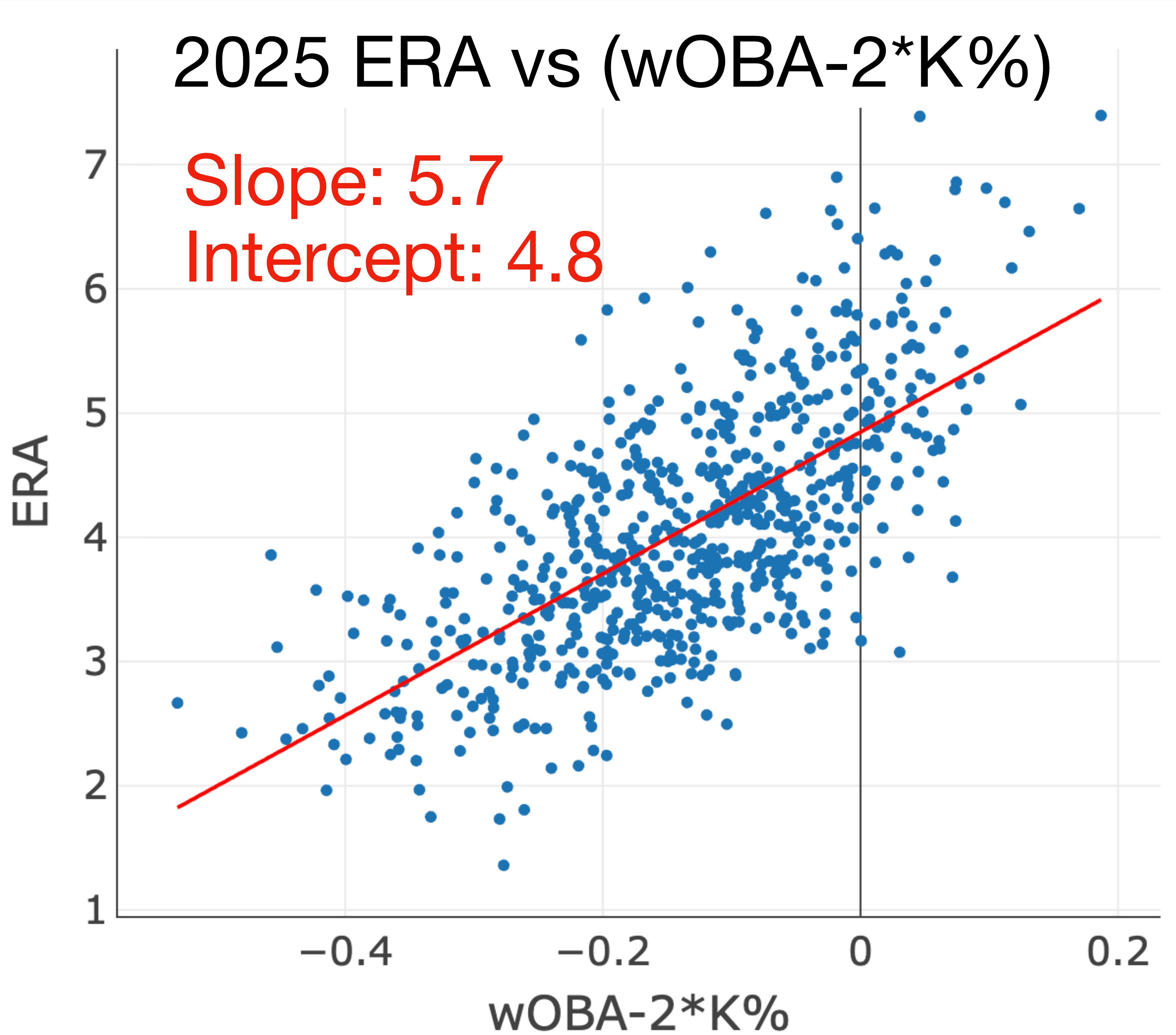
Making a Better Estimator

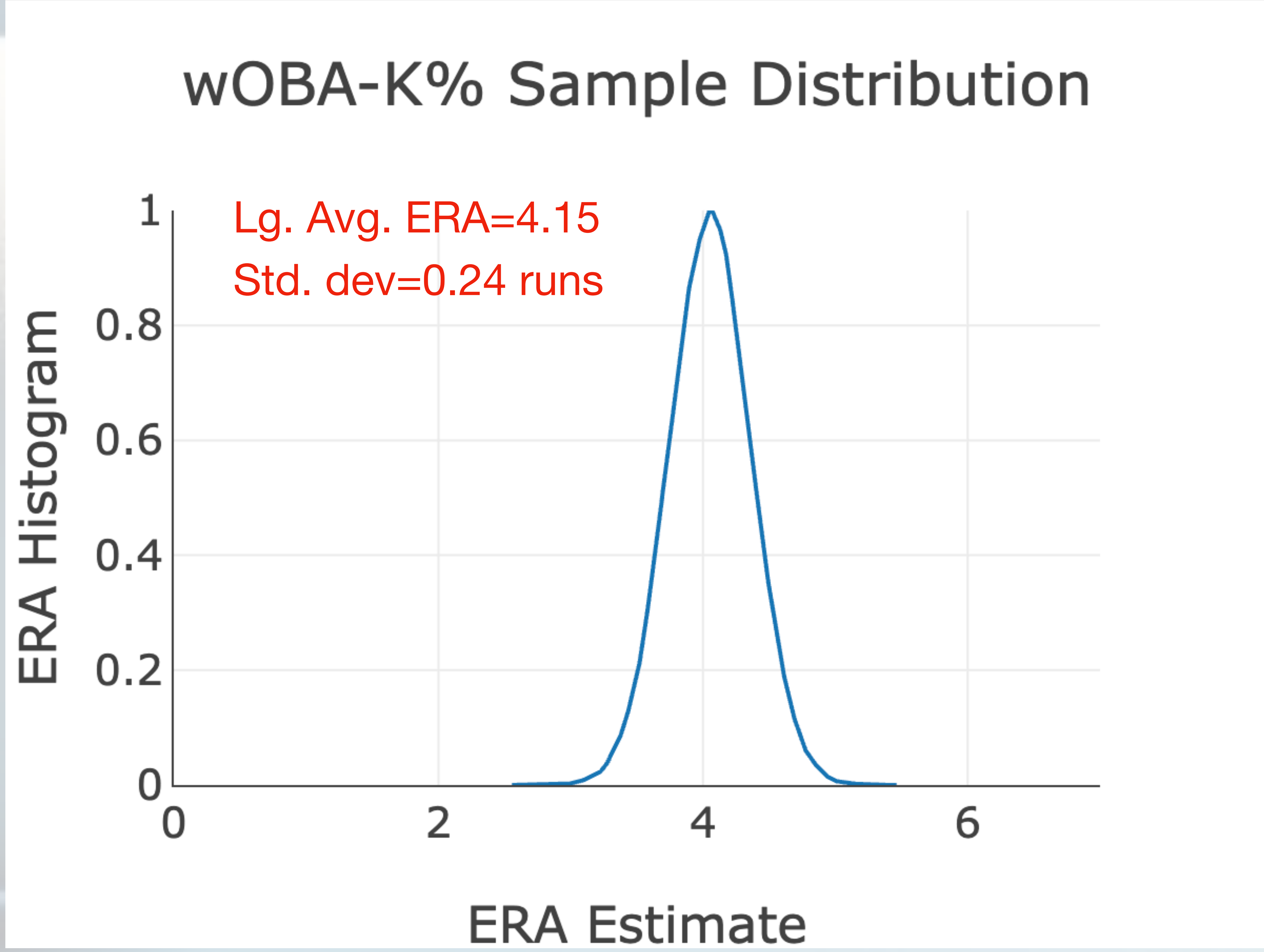
- Estimator variance directly impacts usefulness of the estimator
  - Scaled by the slope of ERA to estimator curve
- One way to increase the slope is subtract the good from the bad: xwOBA-K%

What's Included in Each Estimator

Estimator	K%-BB%	FIP	wOBA
K%			
BB%			
HBP%			
HR%		Too Weighted	
1B, 2B, 3B			

wOBA-2*K%





Correlation coefficient for next year's ERA is 0.42
(better than K%-BB%'s 0.40)

Relation Between Correlation Coefficient and Noise

- The correlation coefficient of ERA plus sampling noise is

$$r = \frac{var\left(ERA - lgERA\right)}{var\left(ERA - lgERA\right) + var(total\ noise)}$$

- Backing out the noise variances

$$var(total\ noise) = var(ERA - lgERA)\frac{1 - r}{r}$$

What Part of the Correlation Coefficient is Due to Sampling Variance?

- For this data, $var(ERA - lgERA) = 0.8$
- A correlation coefficient of 0.4 means that the total variance is

$$var(total\ noise) = var(ERA - lgERA)\frac{1-r}{r} = 0.8\frac{1-0.4}{0.4} = 1.2$$

- Total sampling variance matching a parameter to the next year's ERA is the sum of the sampling errors for the parameter and ERA
  - $var(ERA\ sampling) + var(estimator\ sampling)$= 0.567+0.09=0.657

About half of the overall variance is due to ERA sampling noise!

Future Research

- ERA is a noisy metric, so perhaps we shouldn't be bending over backwards to use it as the matching criterion
- The relationship between wOBA and ERA is nonlinear, so nonlinearities should be included
  - SIERA has them, but need to figure out how those variances also play in
- Balancing the variances and curve fitting might tweak the "best" estimator

Summary

- ERA is very noisy even with “known” underlying stats
  - Fewer samples and much higher variance
- Estimator variance is very important and not explored until now
  - e.g., FIP suffers from high variance due to over-weighting home runs
- xwOBA-K% is a stronger estimator of performance
  - Contains more of the underlying quality of contact than K%-BB%
  - Controls variance better than FIP

Backup Slides

Three True Outcomes (TTO) and FIP

- Three True Outcomes that the pitcher can control are walks, strikeouts, and home runs
  - Independent of defense
- Fielding Independent Pitching (FIP) uses runs created roughly assuming a league average .300 BABIP
- Normalized to look like an ERA (3.20: excellent, 4.20: average, 5.00: awful)

$$FIP = \frac{13 * HR + 3 * (BB + HBP) - 2 * K}{IP} + c$$

$$c = ERA_{lg} - \frac{13 * HR_{lg} + 3 * (BB_{lg} + HBP_{lg}) - 2 * K_{lg}}{IP_{lg}}$$

xFIP vs FIP

- xFIP: FIP using average HR rate per fly ball instead of actual HRs

$$FIP = \frac{13 * \boxed{HR} + 3 * (BB + HBP) - 2 * K}{IP} + c$$

$$xFIP = \frac{13 * \boxed{FlyBalls * \frac{HR_{lg}}{FlyBall}} + 3 * (BB + HBP) - 2 * K}{IP} + FIP_c$$

What is wOBA? (from Zak Payne's talk)

- wOBA was created by Tom Tango and first published in *The Book: Playing the Percentages in Baseball*

- It's a rate stat that assigns an expected run value to each batting event and multiplies it by the number of times that event occurs (weighing certain events higher than others), then divides by PA.

- Once calculated, it's designed to look like OBP; league wOBA is always set to league OBP for easy comparisons (e.g. .400 OBP and .400 wOBA are both considered excellent)

$$wOBA = \frac{0.690 \cdot uBB + 0.722 \cdot HBP + 0.888 \cdot 1B + 1.271 \cdot 2B + 1.616 \cdot 3B + 2.101 \cdot HR}{AB + BB - IBB + SF + HBP}$$

Skill-interactive Earned Run Average (SIERA)

- SIERA quantifies a pitcher's performance by trying to eliminate factors the pitcher can't control by himself
- Unlike xFIP, SIERA adjusts for the type of ball in play.
  - Conceived prior to availability of xwOBA so capture batted balls outcomes differently (by grounders, flyballs and popups)
  - For example, if a pitcher has a high xFIP but has also induced a high proportion of grounders and pop-ups instead of line drives, his SIERA will be lower than his xFIP
- Also includes nonlinear interactions between terms

SIERA Coefficients (as of 2010)

Variable	bpSIERA coefficient	fgSIERA coefficient
(SO/PA)	**-16.986**	**-15.518**
(SO/PA)^2	**7.653**	**9.146**
(BB/PA)	**11.434**	**8.648**
(BB/PA)^2		***27.252***
(netGB/PA)	-1.858	**-2.298**
+/-(netGB/PA)^2	**-6.664**	**-4.920**
(SO/PA)*(BB/PA)		-4.036
(SO/PA)*(netGB/PA)	**10.130**	***5.155***
(BB/PA)*(netGB/PA)	-5.195	4.546
Constant	**6.145**	**5.534**
Year coefficients (versus 2010)		From -0.020 to +0.289
% innings as SP		**0.367**

https://blogs.fangraphs.com/new-siera-part-two-of-five-unlocking-underrated-pitching-skills/

- netGB=(GB-FB), and where ±netGB/PA)^2 is + when GB>FB and – when GB<FB
- Add the following numbers to the constant term for each year:

Year	2002	2003	2004	2005	2006	2007	2008	2009	2010
Coefficient	-0.020	***0.093***	**0.154**	0.037	**0.289**	**0.226**	**0.116**	**0.103**	0.000